

AI Tic-Tac-Toe Using Minimax Algorithm

1. Introduction

Artificial Intelligence can be used to make decisions in games where two players compete against each other.

Tic-Tac-Toe is a simple two-player game played on a 3×3 board. In this project, one player is a human and the other player is an AI.

The AI uses the Minimax algorithm to select its moves.

2. Problem Statement

The objective of this project is to develop a Tic-Tac-Toe game in which the computer can play against a human player.

The AI should be able to examine the available moves and select a move that gives it the best possible outcome.

3. Objectives

The main objectives are:

1. To implement a command-line Tic-Tac-Toe game.
2. To understand game states in Artificial Intelligence.
3. To implement the Minimax algorithm.
4. To allow a human player to compete against the AI.
5. To understand how AI makes decisions in a two-player game.

4. AI Concept

Minimax Algorithm

Minimax is an algorithm used for decision making in competitive games.

It assumes that both players try to make the best possible move.

The AI tries to maximize its score, while the opponent tries to minimize the AI's score.

For this project, the scores are:

Result	Score
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Result	Score
AI wins	1
Draw	0
Player wins	-1

The algorithm explores the possible moves and returns a score for each possible situation.

5. Working of the Project

The program first creates an empty 3×3 board.

The human player makes the first move using x.

After the human move, the program checks whether the game has ended.

If the game is still continuing, the AI uses Minimax to examine the available moves.

The AI places o in the selected position.

This process continues until either:

- The player wins.
- The AI wins.
- The board becomes full and the game is a draw.

6. Algorithm

1. Start the game.
2. Create an empty Tic-Tac-Toe board.
3. Display the board.
4. Take the player's move.
5. Check whether the move ends the game.
6. If the game is not over, calculate the AI move using Minimax.
7. Place o on the selected position.
8. Check the game result.
9. Repeat the process until the game ends.
10. Display the final result.

7. Implementation

The project is implemented using Python.

The main functions are:

- `print_board()` — displays the board.
- `check_winner()` — checks for a winner or draw.
- `minimax()` — calculates the score of possible game states.
- `find_best_move()` — selects the best move for the AI.
- `play_game()` — controls the complete game.

8. Input and Output

Input

The user enters a number from 1 to 9 representing the position where x should be placed.

Output

The program displays the updated board and the move selected by the AI.

Example:

Enter your move (1-9): 5

AI is thinking...

```

  O |   |
  ---+---+---
      | X |
  ---+---+---
      |   |

```

9. Advantages

- Simple and easy to use.
- Does not require a graphical interface.
- Does not require external libraries.
- Demonstrates an important AI game-playing algorithm.
- Can be executed directly from a terminal.

10. Limitations

The project is limited to a standard 3×3 Tic-Tac-Toe board.

The current version does not include a graphical interface or different difficulty levels.

11. Future Scope

The project can be improved by adding:

- Easy, medium, and hard difficulty levels.
- A graphical user interface.
- Player score tracking.
- Sound effects.
- Larger game boards.
- Other game-playing algorithms.

12. Conclusion

The project demonstrates how Artificial Intelligence can be applied to a simple game.

By using the Minimax algorithm, the AI evaluates possible game states and selects a suitable move.

This project provides a basic understanding of adversarial search and decision making in Artificial Intelligence.